

In this study we investigated the genes and pathways affected by the combination therapy of RITA (a p53 activator) and CPI-203 (a BET inhibitor) in acute myeloid leukaemia (AML) by comparing three AML stem cells from three treated and three untreated patients. We first assessed data quality by comparing the distribution of gene expression across treated and untreated samples using density plots (**Fig. 1A**), and by visualising the relationship between mean expression levels and $\log_2$ fold changes of up- and down-regulated genes using an MA plot (**Fig. 1B**). All these quality-assessment techniques indicated reliable data. However, additional replicates would have improved statistical power, particularly because one untreated sample showed an unusual number of peaks and some genes had mean expression levels below 100. Principal component analysis (PCA) further revealed a clear difference between treated and untreated AML stem cells along the first principal component that explained 43% of variance indicating treatment-induced transcriptomic changes (**Fig. 1C**).

We detected 1184 downregulated genes and 570 upregulated genes by using a fold-change threshold of greater less than -1 and greater than 1 respectively, demonstrating a substantial transcriptomic alteration produced by the treatment (**Fig. 1D**). Moreover, the expression of the top five downregulated genes is known to support growth and protecting AML cells from apoptosis. For example, the gene CD84 and hepatocyte growth factor (HGF)'s role in apoptosis have been demonstrated in knock out experiments (1–3). The upregulated genes, like TXNIP among the top five, have been associated with AML cell arrest and apoptosis when upregulated (4) (**Fig. 1D**). The heatmap further also showed that the treatment induced a stronger pattern of gene downregulation than upregulation (**Fig. 1E**). Furthermore, of the 10 significantly differentially expressed genes, 90% were downregulated in AML cells treated with the combination of RITA and CPI-203 (**Fig. 1F**). This may indicate a synergistic therapeutic effect that is largely mediated by the downregulated gene set.

To investigate pathways affected by gene downregulation, we performed an over-representation analysis (ORA) of Gene Ontology biological processes. Downregulated genes were primarily enriched in pathways related to myeloid immune cell interaction, activation, and proliferation. Among these, *leukocyte cell–cell adhesion* was the most significantly enriched GO term (adjusted $p < 0.01$), with the highest number of overlapping genes (**Fig. 2A, Fig. 2B**). These findings suggest that the treatment downregulates leukocyte cell–cell adhesion, a pathway known to support AML cell proliferation, survival and chemotherapy resistance (5). We examined gene expression within the *leukocyte cell–cell adhesion* GO term which revealed a downregulation pattern of genes (**Fig. 2C**), consistent with Hultmark's report that CPI-230 primes for death and downregulates genes involved in this pathway (6). This pathway comprised 50 significantly downregulated genes involved in cell adhesion and proliferative signalling, including *PECAM1*, *SELL*, *CCR2*, and *CD31*. Notably, downregulation of *CCR2* has been shown to limit AML cell proliferation through disruption of the CCL2–CCR2 axis (7) (**Fig. 2D**). Furthermore, we did a STRING analysis that revealed interaction networks among several downregulated genes, including *CD80*, *CD4*, *CCR2*, and *PECAM1*, indicating coordinated suppression of adhesion-associated signalling pathways that support AML cell proliferation and survival following treatment (**Fig. 2E**).

We next performed an over-representation analysis (ORA) of upregulated genes, which revealed significant enrichment of pathways associated with cellular stress responses and regulation of cell death (**Fig. 3A, Fig. 3B**). The top enriched GO term was *response to radiation*, consistent with the known mechanism of RITA, which activates p53 target genes also induced by radiation-mediated apoptosis (8–11). This finding indicates that the treatment effectively activates stress-induced and apoptotic signalling pathways in AML cells. To further investigate this pathway, we mapped the genes associated with the *response to radiation* GO term and observed consistent upregulation in the treated group (**Fig. 3C**). To determine whether these genes were bona fide p53 targets, we visualised the expression of the 32 overlapping genes using a box plot. Several of these genes, including *CDKN1A*, *GADD45A*, and *EGR1*, are well-established p53 target genes involved in apoptosis and cell-cycle arrest (12,13) (**Fig. 3D**). Finally, within this pathway, we used STRING to identify existing gene–gene interactions among the 32 overlapping genes. This analysis revealed interactions between *EGR1* and *SIRT1*, *DUSP1*, and *GADD45A*, all of which are implicated in cell-cycle arrest (Fig. 3E).

Overall, although more genes were downregulated in this experiment, the overrepresentation analysis (ORA) of all significant genes identified *cell-to-cell adhesion* as the most enriched pathway. This pathway was also the top enriched term in the ORA of upregulated genes (Fig. 3F). Importantly, this result reflects the combined effects of the treatment rather than the effect of an individual drug.

The analysis further indicates that the affected genes and pathways are predominantly associated with cell death–related outcomes. However, this conclusion would be strengthened by functional validation assays of the key genes, which could also help explain why the enriched pathways among upregulated genes appear more prominent despite the larger number of downregulated genes.

One limitation of this analysis is the absence of a cellular component Gene Ontology discussion, which could have provided additional insight into the cellular structures and processes underlying the observed biological effects. In summary, the results suggest that the combination therapy portrays its mechanism of action in AML by suppressing cellular growth and inducing apoptosis, a pattern that has also been reported in CML by Abraham *et al.* (14).

References

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